

APS110 Tutorial

#10

Daniel Brlas

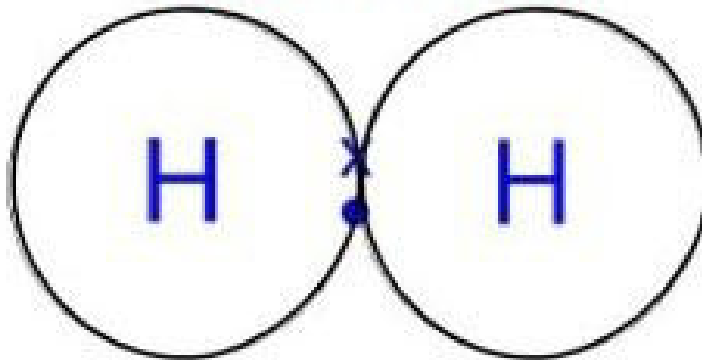
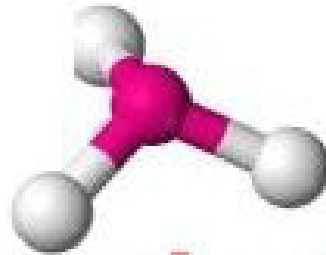
Email: daniel.brlas@mail.utoronto.ca

Review – Bonding

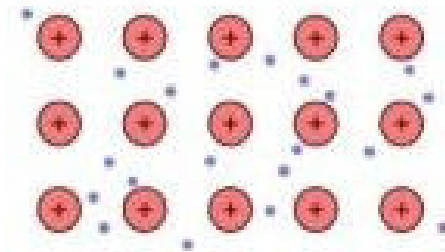
Ionic



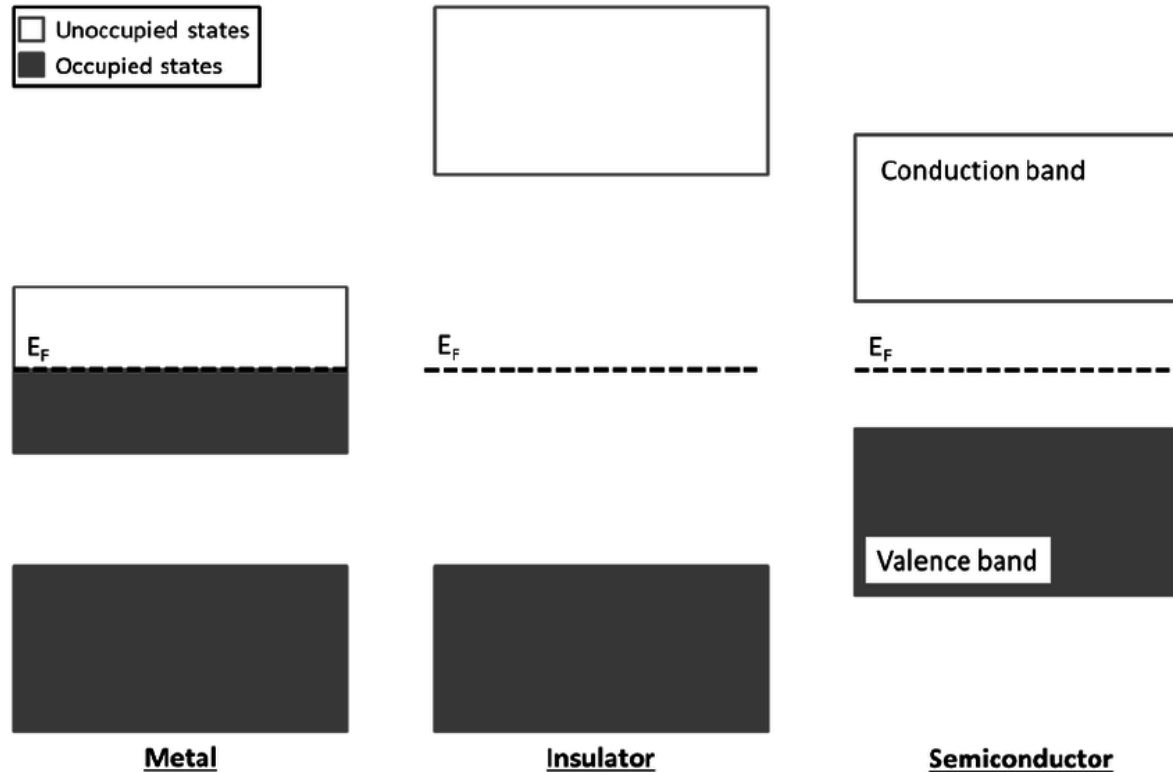
Covalent



Metallic

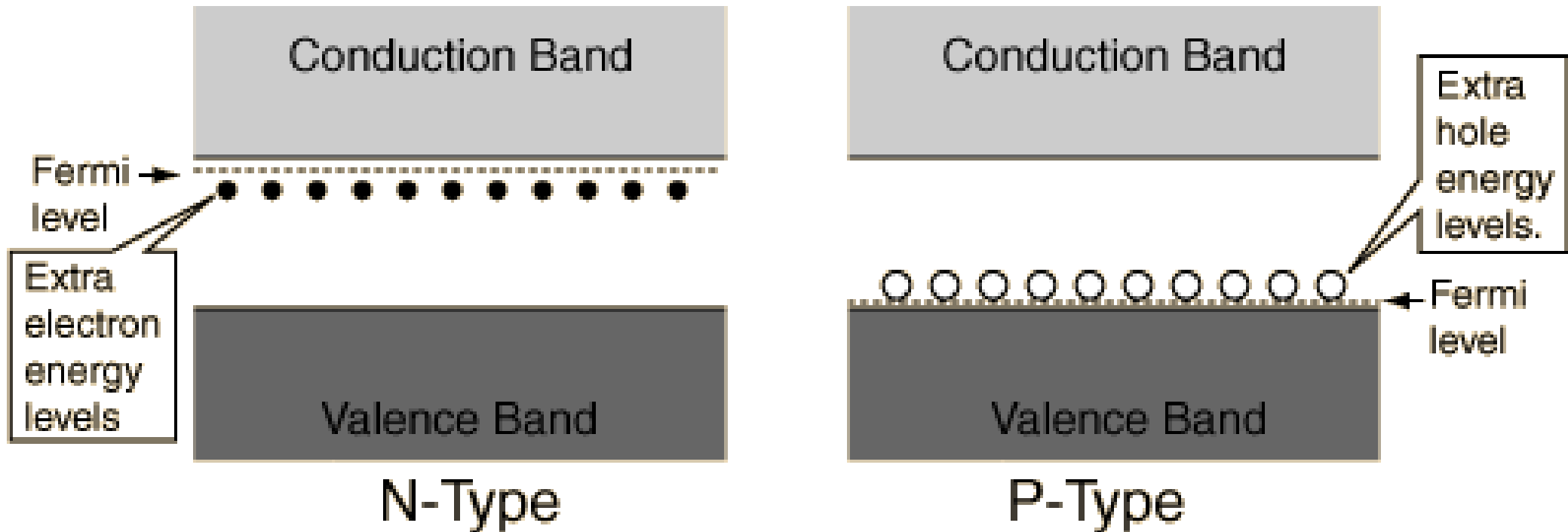


Review – Band Theory

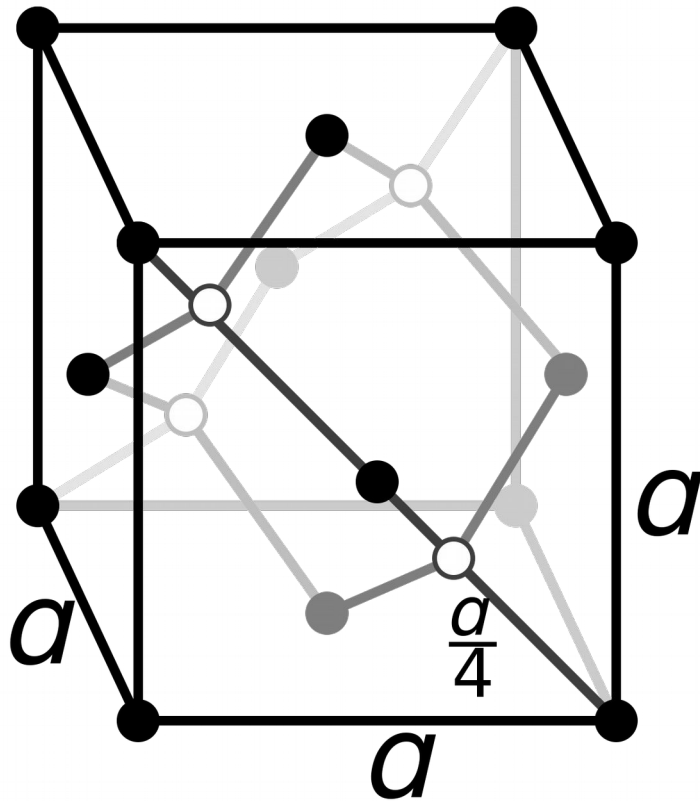


<https://www.allaboutcircuits.com/textbook/semiconductors/chpt-2/band-theory-of-solids/>

Review – Semiconductor Types

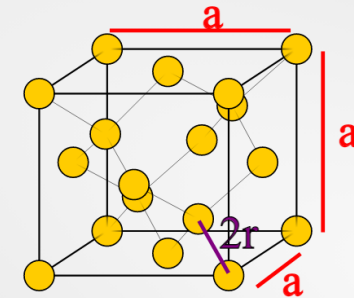


Review – Diamond Cubic



DC Visual Representation with Open Space	Hard Sphere Model--Atoms Touch in the purple direction Cube side length a shown in blue	Where atoms touch, the distance is $2r$	The atom coordinates are $(0,0,0)$ and $(1/4, 1/4, 1/4)$ in relation to the cube side length, a.
		$2r = \sqrt{\left(\frac{1}{4}a\right)^2 + \left(\frac{1}{4}a\right)^2 + \left(\frac{1}{4}a\right)^2}$ $2r = \sqrt{3} \left(\frac{1}{4}a\right)$	3D Pythagorean Theorem (You can also use multiple 2D Pythagorean Theorems)
Lattice Parameter $a = \frac{8}{\sqrt{3}}r$			

Review – Diamond Cubic



Diamond Cubic
Open Visualization

$$a = \frac{8}{\sqrt{3}}r$$

$$\text{Volume} = a^3$$

8 atoms on each corner,
with 1/8 of its volume
inside the unit cell (1)

+ 6 atoms on each face
with 1/2 of its volume
inside the unit cell (3)

+ 4 atoms inside
with 100% of its volume
inside the unit cell (4)

Volume of atoms =

$$8 \cdot \frac{4}{3}\pi r^3 = 8 \text{ atoms per unit cell}$$

Volume of cube = $\left(\frac{8}{\sqrt{3}}r\right)^3$

$$\text{APF} = \frac{\text{Volume of atoms}}{\text{Volume of unit cell}}$$

For DC

$$\text{APF} = \frac{\sqrt{3}\pi}{16} = 34\%$$

Midterm Review

Practice Question

1. At room temperature, the electrical conductivity and electron mobility for aluminum are $3.8 \times 10^7 \text{ S/m}$ [or $(\Omega \cdot \text{m})^{-1}$] and $0.0012 \text{ m}^2/\text{V} \cdot \text{s}$, respectively.
 - a) Compute the number of free electrons per cubic meter for aluminum at room temperature.
 - b) What is the number of free electrons per aluminum atom? Assume a density of 2.7 g/cm^3 .

Practice Question

a)

$$\sigma = ne^{-}\mu_e \quad \text{only asking for electrons}$$

$$n = \frac{\sigma}{e^{-}\mu_e} = \frac{3.8 \times 10^7}{(1.602 \times 10^{-19}) \cdot (0.0012)} = 1.98 \times 10^{29} \text{ m}^{-3}$$

b)

b)

$$N_{Al} = \frac{N_A \cdot \rho}{A_{Al}} = \frac{(6.022 \times 10^{23}) \cdot (2.71) \cdot \left(\frac{100 \text{ cm}}{1 \text{ m}}\right)^3}{26.98} = 6.03 \times 10^{28} \text{ m}^{-3}$$

$$\frac{n}{N_{Al}} = \frac{1.98 \times 10^{29} \text{ m}^{-3}}{6.03 \times 10^{28} \text{ m}^{-3}} = 3.28$$

Practice Quiz

1. Which of the following would be most likely to cause cross-linking of a polymer?
 - a) Radio frequency radiation
 - b) Microwave radiation
 - c) X-ray radiation
 - d) Gamma radiation □ highest energy
2. Which of the following is the correct electronic configuration for a neutral atom of Sc^{3+} ?
 - a) $[\text{Ar}]$
 - b) $[\text{Kr}] 4d^1$
 - c) $[\text{Kr}] 3d^1$
 - d) $[\text{Kr}] 3d^1 4s^1$
3. What type of bonding is present in carbon in the diamond cubic crystal structure?
 - a) Ionic
 - b) Covalent
 - c) Metallic
 - d) Mixed ionic and covalent

Practice Quiz

4. How many atoms are present within a single unit cell of the diamond cubic crystal structure?
 - a) 4
 - b) 5
 - c) 6
 - d) 8
5. A non-metallic material has a bandgap of 2.2 eV. In terms of which wavelength of light that passes through the material, which of the following is true?
 - a) The material passes all wavelengths of visible light
 - b) The material does not pass any wavelengths of visible light (i.e. it is opaque)
 - c) The material passes red light but not blue light
 - d) The material passes blue light but not red light
6. The following characteristics for both intrinsic and p-type extrinsic GaSb at room temperature. Calculate the hole and electron mobility based on this data

	σ ($\Omega \cdot \text{m}^{-1}$)	n (m^{-3})	p (m^{-3})
Intrinsic	2.5×10^{-6}	3.0×10^{14}	3.0×10^{13}
Extrinsic (p-type)	3.6×10^{-5}	4.5×10^{13}	2.0×10^{12}

Intrinsic)

$$\sigma = ne^- \mu_e + ph^+ \mu_h$$
$$2.5 \times 10^{-6} = (3.0 \times 10^{14})(1.602 \times 10^{-19})\mu_e + (3.0 \times 10^{14})(1.602 \times 10^{-19})\mu_h$$
$$0.52 = \mu_e + \mu_h$$

Extrinsic)

Extrinsic)

$$3.6 \times 10^{-5} = (4.5 \times 10^{14})(1.602 \times 10^{-19})\mu_e + (2.0 \times 10^{12})(1.602 \times 10^{-19})\mu_h$$
$$112.4 = 225\mu_e + \mu_h \Rightarrow \mu_h = 112.4 - 225\mu_e$$

$$0.52 = \mu_e + (112.4 - 225\mu_e) \Rightarrow \mu_e = 0.5 \frac{m^2}{V \cdot s}, \mu_h = 0.02 \frac{m^2}{V \cdot s}$$